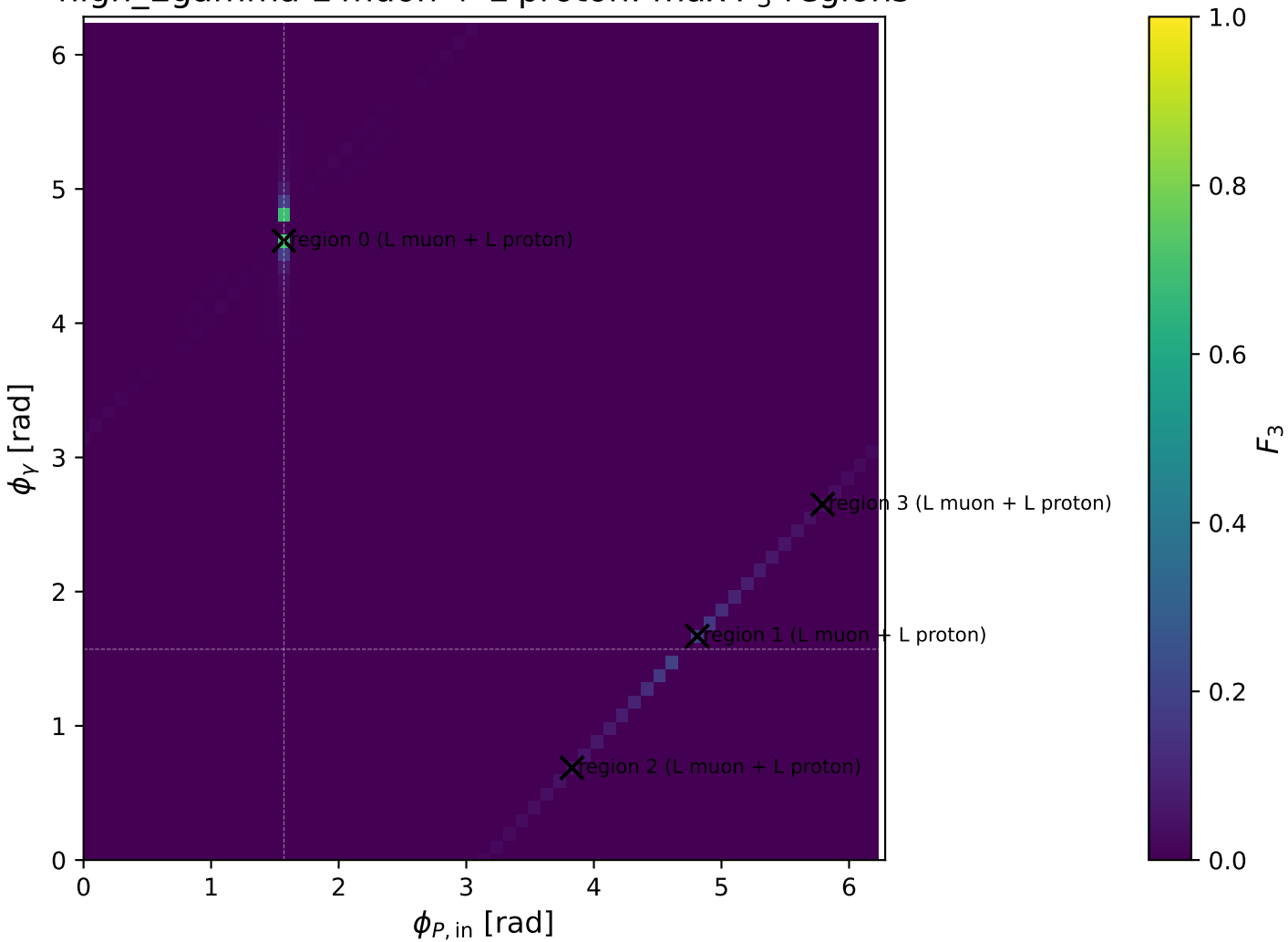
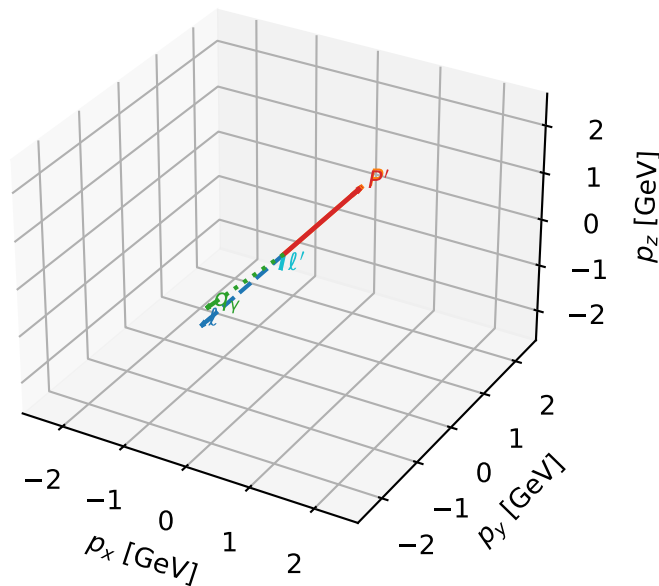


high\_Egamma L muon + L proton: max  $F_3$  regions



# L muon + L proton characteristic max $F_3$ configuration

3D momenta

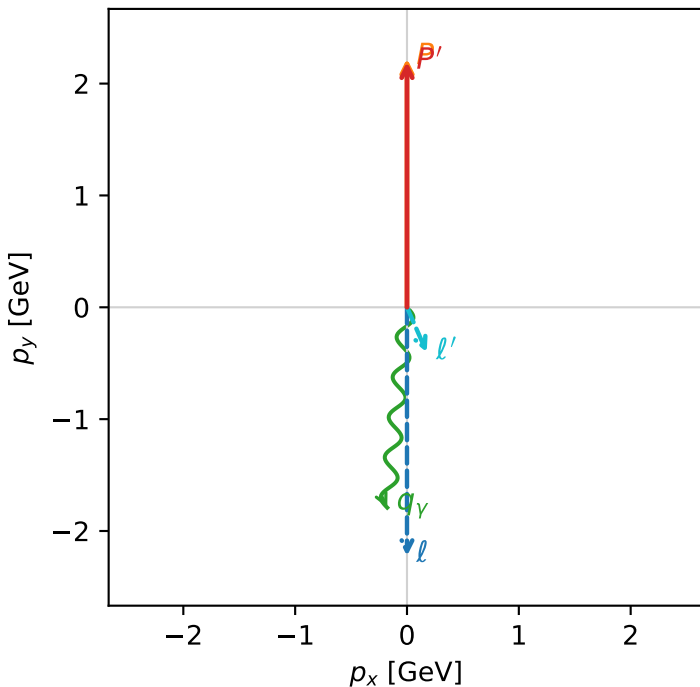


f3\_high\_egamma\_ll\_region\_0 F\_3=0.685592  
 region: high\_Egamma\_LL\_region\_0  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|L_\mu\rangle \otimes |L_p\rangle$

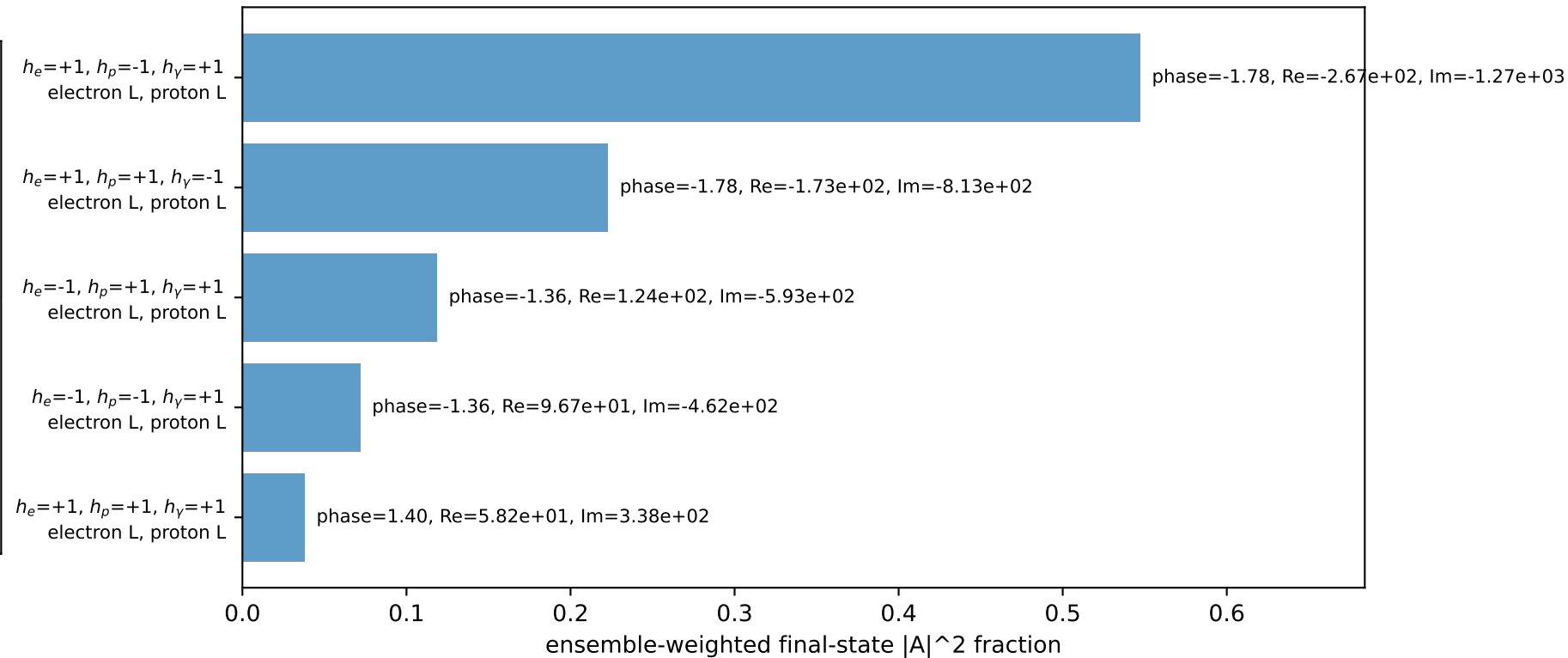
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=2.19415$   
 $\theta_{in}=1.57079$ ,  $\phi_l=4.71239$ ,  $\phi_p=1.5708$   
 $E_Y=1.8$ ,  $\phi_Y=4.61421$   
 $Q^2=0.199845$ ,  $x_B=0.0120018$ ,  $t=-0.000128163$ ,  $W^2=17.3312$ ,  $y=0.807338$   
 $\theta(l', \gamma)=29.278$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= -0.0000$   $p_y= -2.2231$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 0.0000$   $p_y= 2.2231$   $p_z= 0.0000$   
 $l'$   $E= 0.4523$   $p_x= 0.1764$   $p_y= -0.4028$   $p_z= 0.0000$   
 $P'$   $E= 2.3862$   $p_x= 0.0000$   $p_y= 2.1942$   $p_z= 0.0000$   
 $q_Y$   $E= 1.8000$   $p_x= -0.1764$   $p_y= -1.7913$   $p_z= 0.0000$

Transverse plane

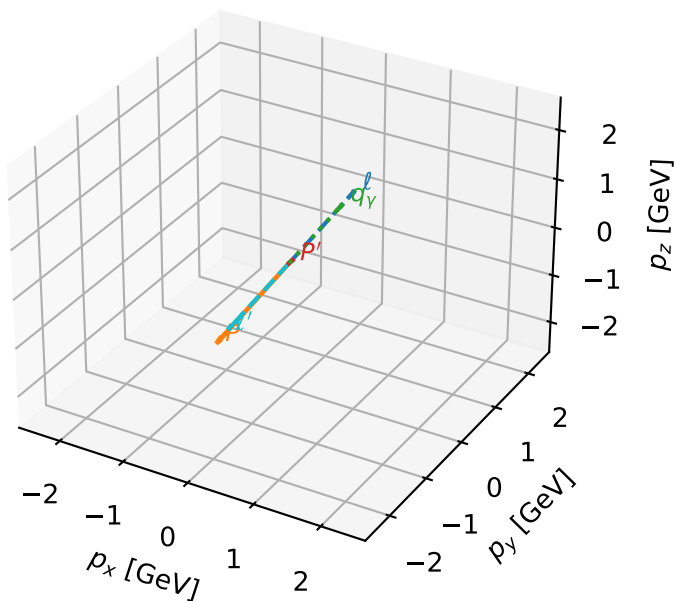


Leading initial-ensemble/final-state components (N=5, retained=99.9%)



# L muon + L proton characteristic max $F_3$ configuration

## 3D momenta

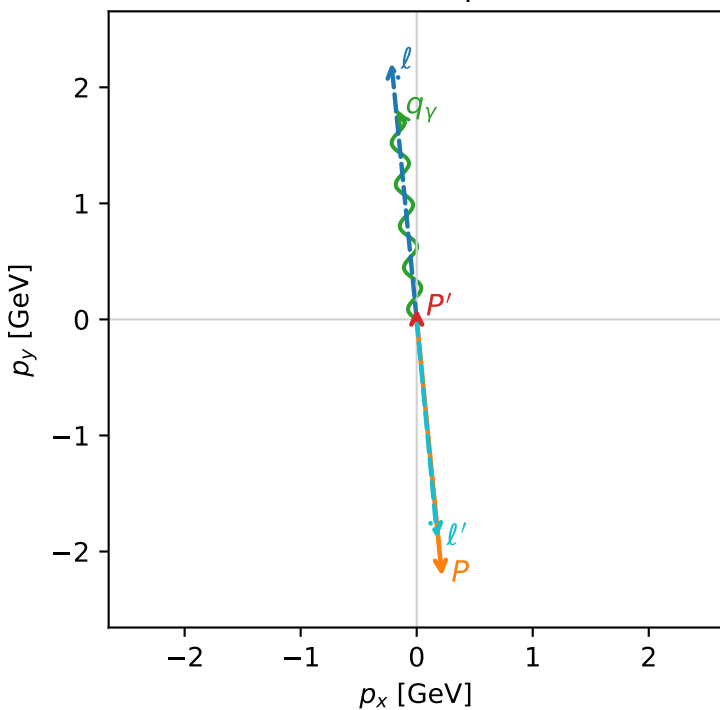


f3\_high\_egamma\_ll\_region\_1 F\_3=0.195276  
 region: high\_Egamma\_LL\_region\_1  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=0.0933634$   
 $\theta_{in}=1.57079$ ,  $\phi_l=1.66897$ ,  $\phi_p=4.81056$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=1.66897$   
 $Q^2=16.833$ ,  $x_B=0.84622$ ,  $t=-3.20239$ ,  $W^2=3.93884$ ,  $y=0.964469$   
 $\theta(l', \gamma)=179.723$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= -0.2179$   $p_y= 2.2124$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 0.2179$   $p_y= -2.2124$   $p_z= 0.0000$   
 $l'$   $E= 1.8959$   $p_x= 0.1764$   $p_y= -1.8847$   $p_z= 0.0000$   
 $P'$   $E= 0.9426$   $p_x= 0.0000$   $p_y= 0.0934$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -0.1764$   $p_y= 1.7913$   $p_z= 0.0000$

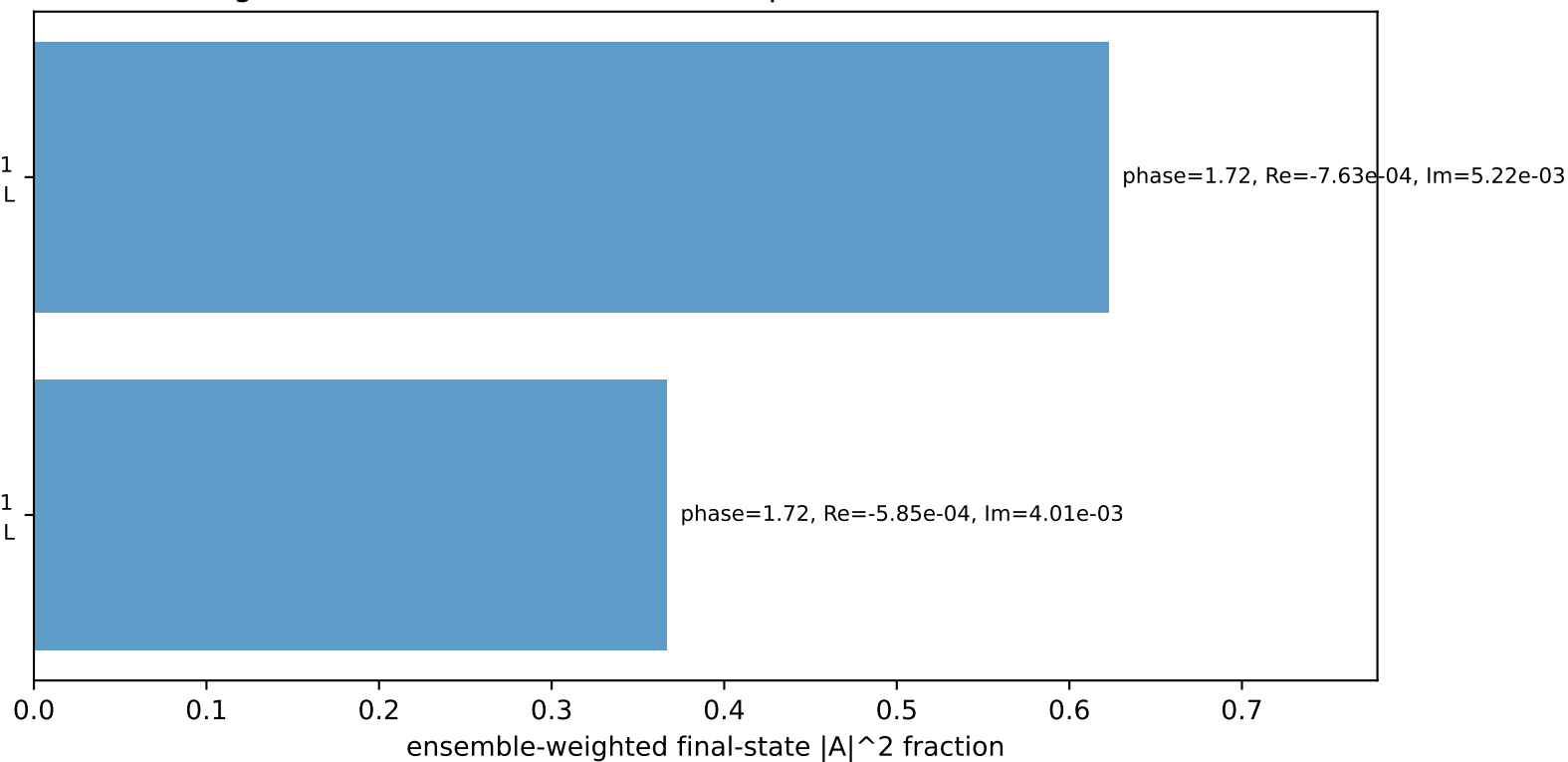
## Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$   
 electron L, proton L

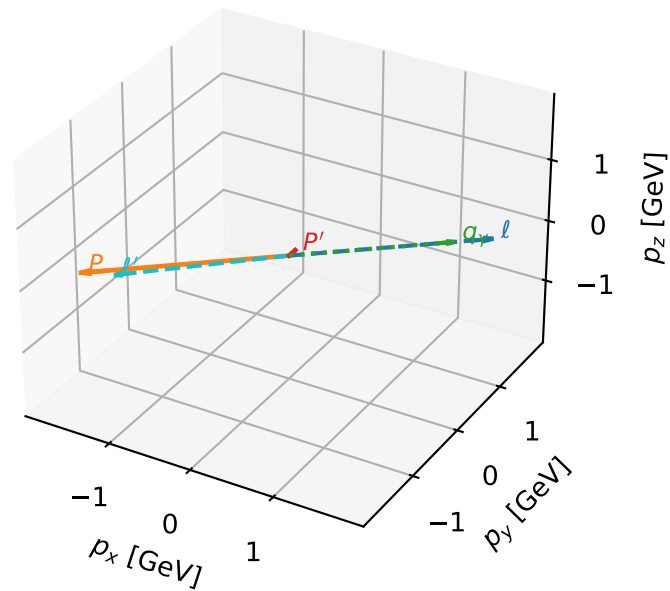
$h_e=-1, h_p=+1, h_\gamma=-1$   
 electron L, proton L

## Leading initial-ensemble/final-state components (N=2, retained=99.0%)



# L muon + L proton characteristic max $F_3$ configuration

3D momenta



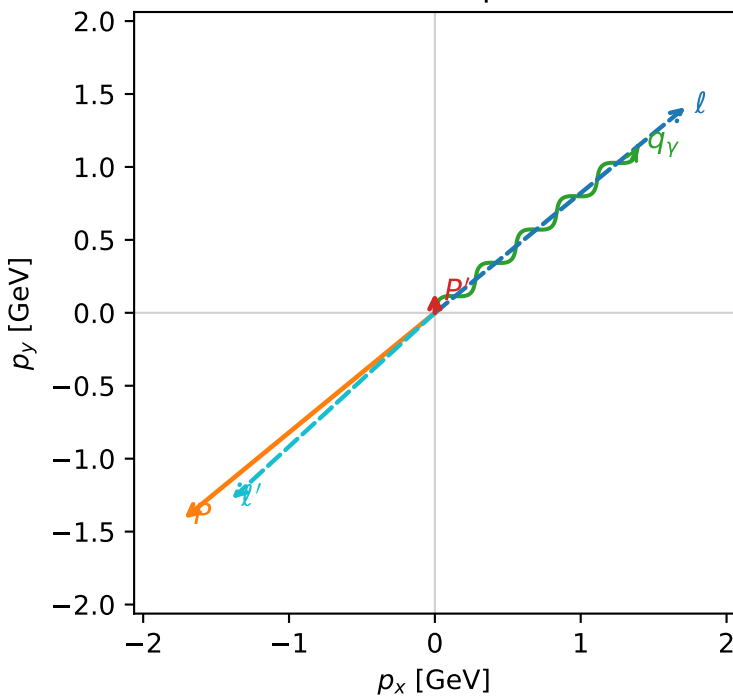
f3\_high\_egamma\_ll\_region\_2  $F_3=0.0475577$   
 region: high\_Egamma\_LL\_region\_2  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=0.13423$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=0.687223$ ,  $\phi_p=3.82882$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=0.687223$   
 $Q^2=16.7766$ ,  $x_B=0.843841$ ,  $t=-3.19163$ ,  $W^2=3.98449$ ,  $y=0.963948$   
 $\theta(\ell', \gamma)=176.85$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:

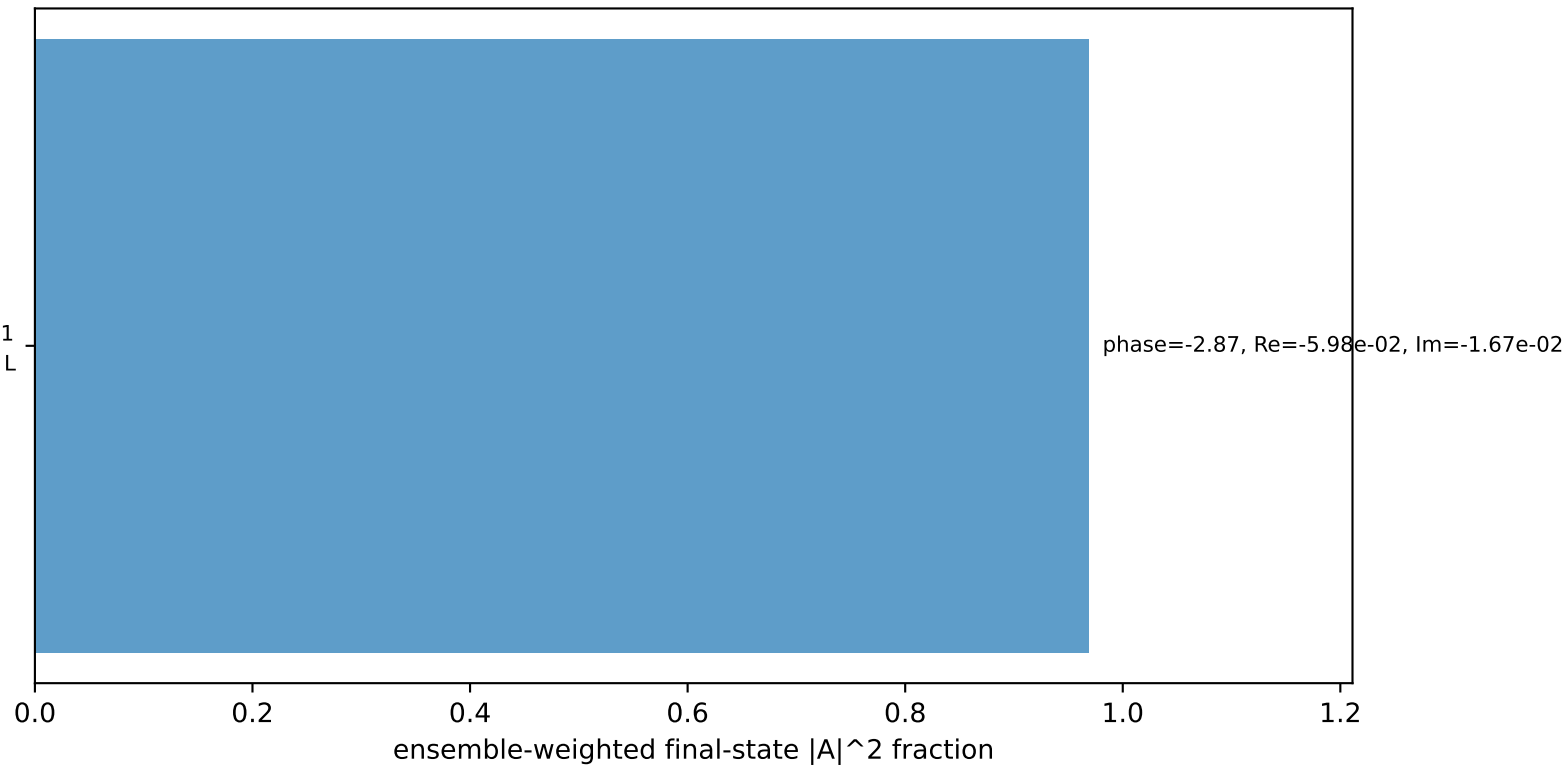
$\ell$   $E= 2.2256$   $p_x= 1.7185$   $p_y= 1.4103$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= -1.7185$   $p_y= -1.4103$   $p_z= 0.0000$   
 $\ell'$   $E= 1.8910$   $p_x= -1.3914$   $p_y= -1.2761$   $p_z= 0.0000$   
 $P'$   $E= 0.9476$   $p_x= 0.0000$   $p_y= 0.1342$   $p_z= 0.0000$   
 $q_Y$   $E= 1.8000$   $p_x= 1.3914$   $p_y= 1.1419$   $p_z= 0.0000$

Transverse plane



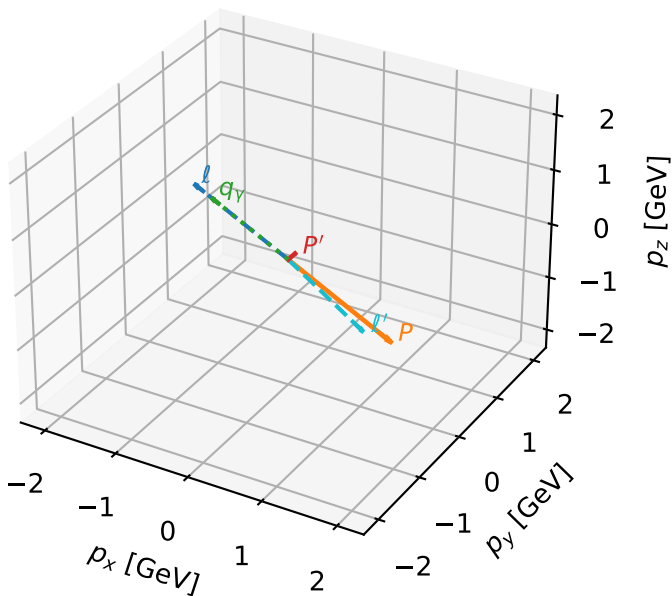
$h_e=-1$ ,  $h_p=-1$ ,  $h_\gamma=+1$   
 electron L, proton L

Leading initial-ensemble/final-state components (N=1, retained=96.9%)



# L muon + L proton characteristic max $F_3$ configuration

3D momenta

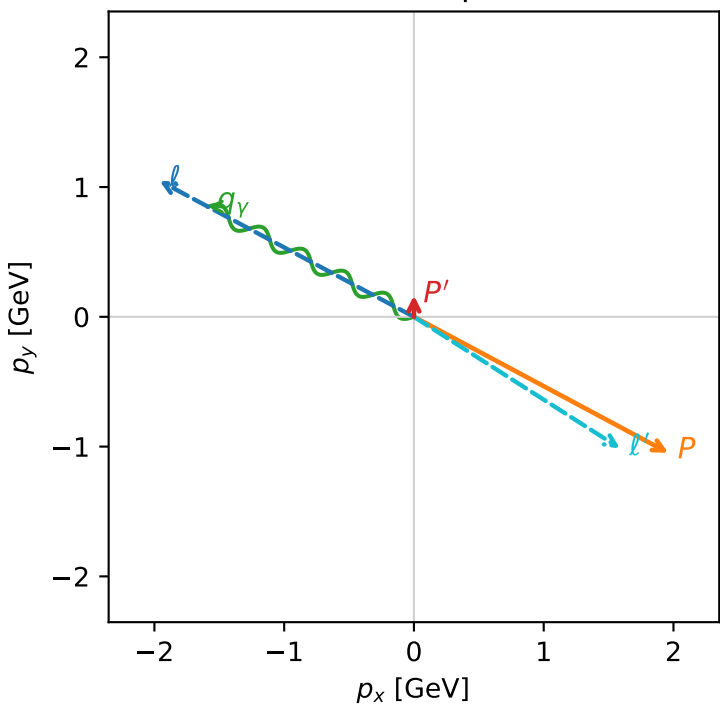


f3\_high\_egamma\_ll\_region\_3 F\_3=0.0388992  
region: high\_Egamma\_LL\_region\_3  
outgoing-state purity: 1  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22311$ ,  $|\vec{P}'|=0.16464$   
 $\theta_{in}=1.57079$ ,  $\phi_l=2.65072$ ,  $\phi_p=5.79231$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=2.65072$   
 $Q^2=16.7218$ ,  $x_B=0.841525$ ,  $t=-3.18118$ ,  $W^2=4.02887$ ,  $y=0.963441$   
 $\theta(l', \gamma)=175.578$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2256$   $p_x= -1.9606$   $p_y= 1.0480$   $p_z= -0.0000$   
 $P$   $E= 2.4129$   $p_x= 1.9606$   $p_y= -1.0480$   $p_z= 0.0000$   
 $l'$   $E= 1.8862$   $p_x= 1.5875$   $p_y= -1.0132$   $p_z= 0.0000$   
 $P'$   $E= 0.9523$   $p_x= 0.0000$   $p_y= 0.1646$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -1.5875$   $p_y= 0.8485$   $p_z= 0.0000$

Transverse plane



$h_e=-1$ ,  $h_p=-1$ ,  $h_\gamma=+1$   
electron L, proton L

Leading initial-ensemble/final-state components (N=1, retained=97.7%)

